

Machine Learning for Predicting Supplemental Aflibercept

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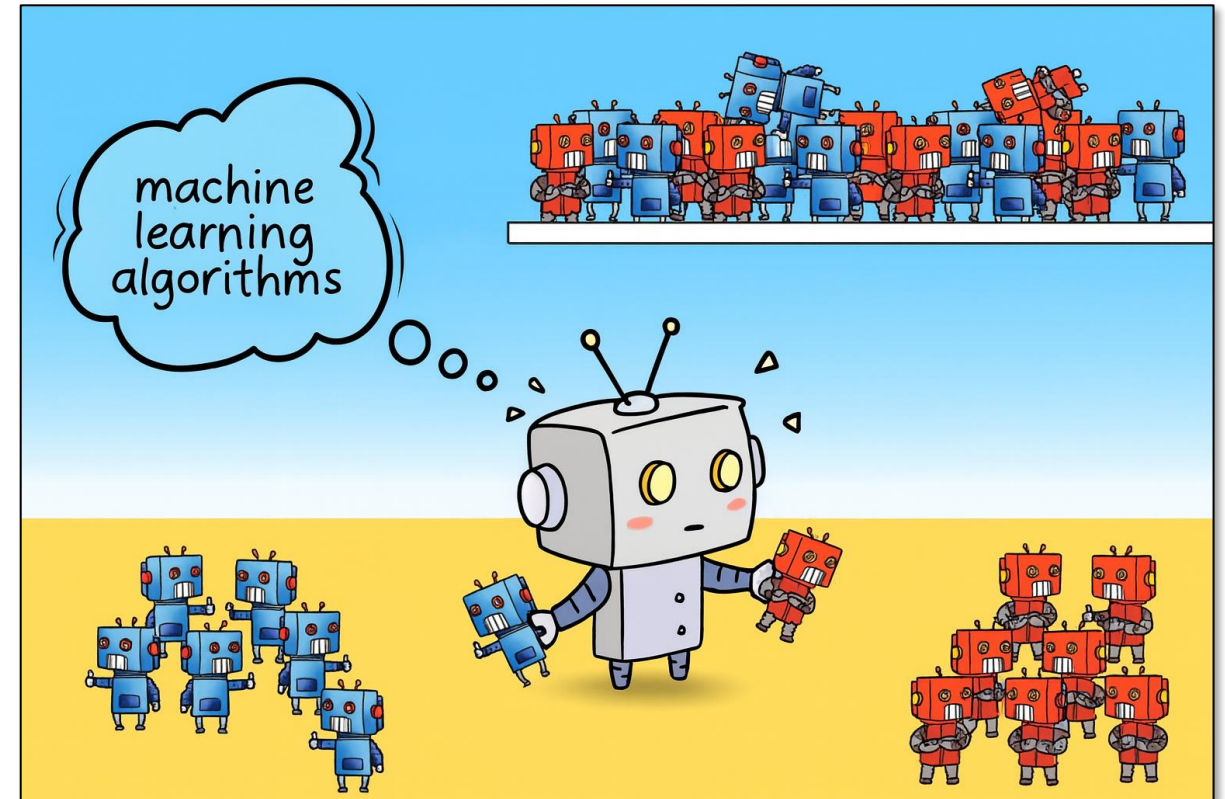
ADVERUM

Agenda

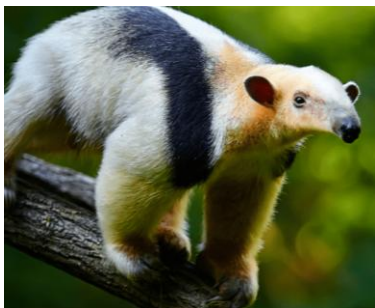
1. What is Machine Learning?
2. Purpose
3. Project Steps
4. Results
5. Limitations/Future Steps
6. Thank You

Basic Definition

- Programming computers to learn and improve from experience
- Typically involves finding patterns in data to make predictions or decisions
- Statistics is about inference and understanding relationships, while Machine Learning is about prediction + classification and optimizing performance



Anteater



Duck



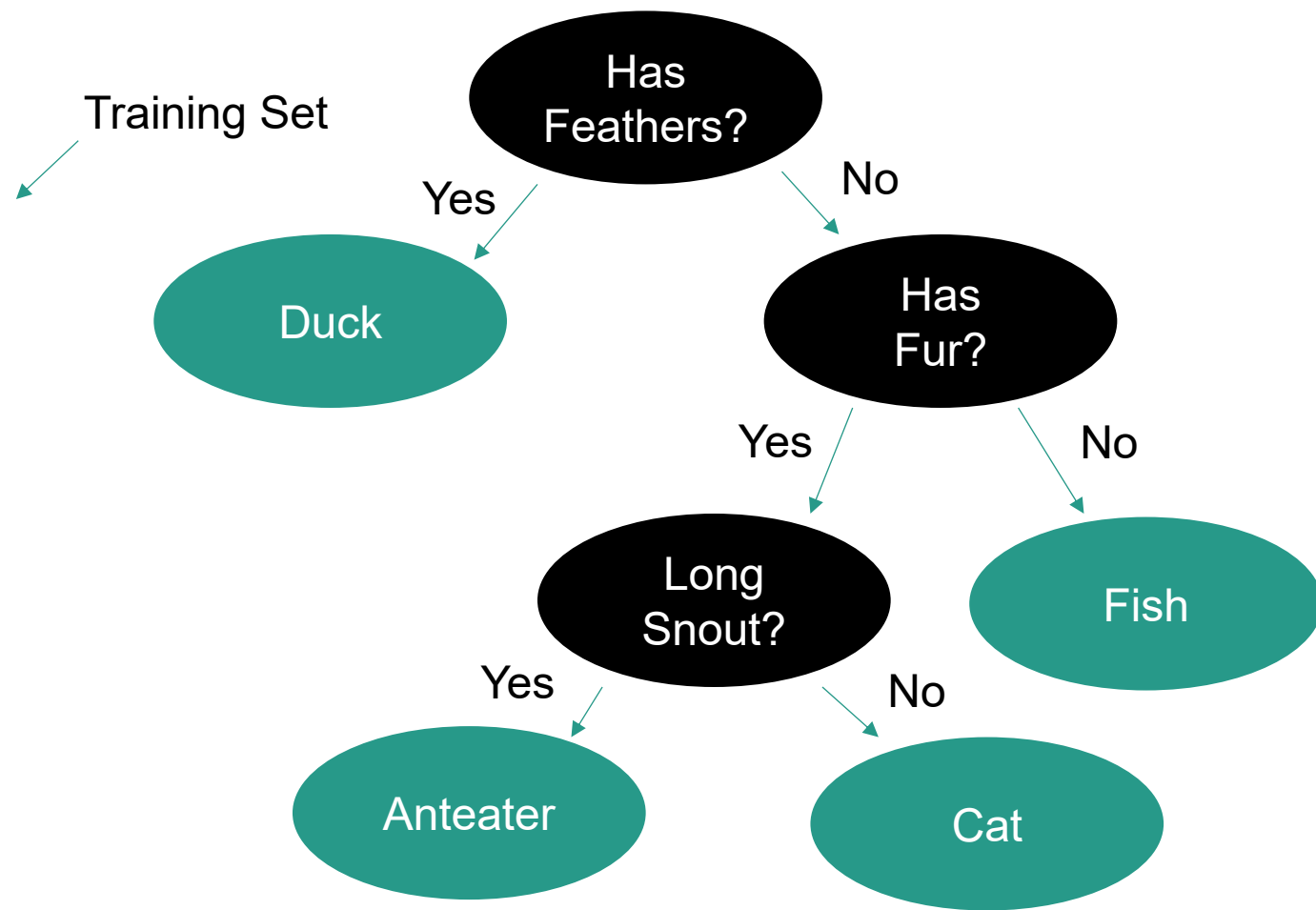
Fish



Cat

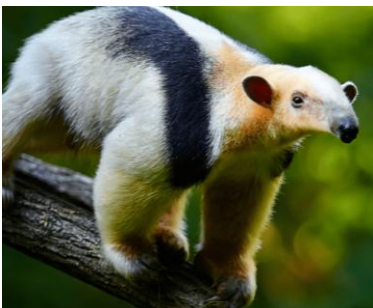


Training Set



Testing Set

Anteater



Duck



Fish

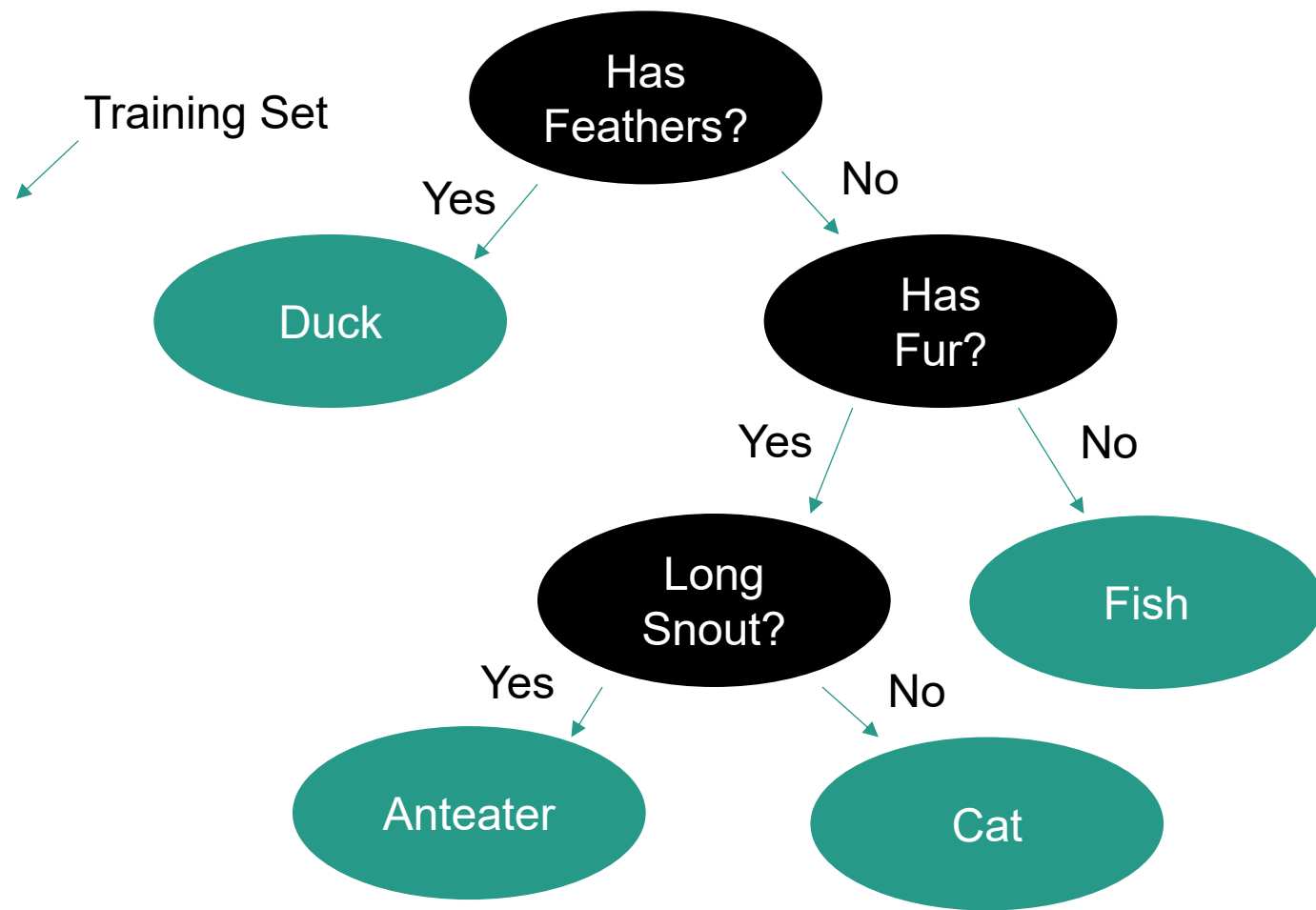


Cat



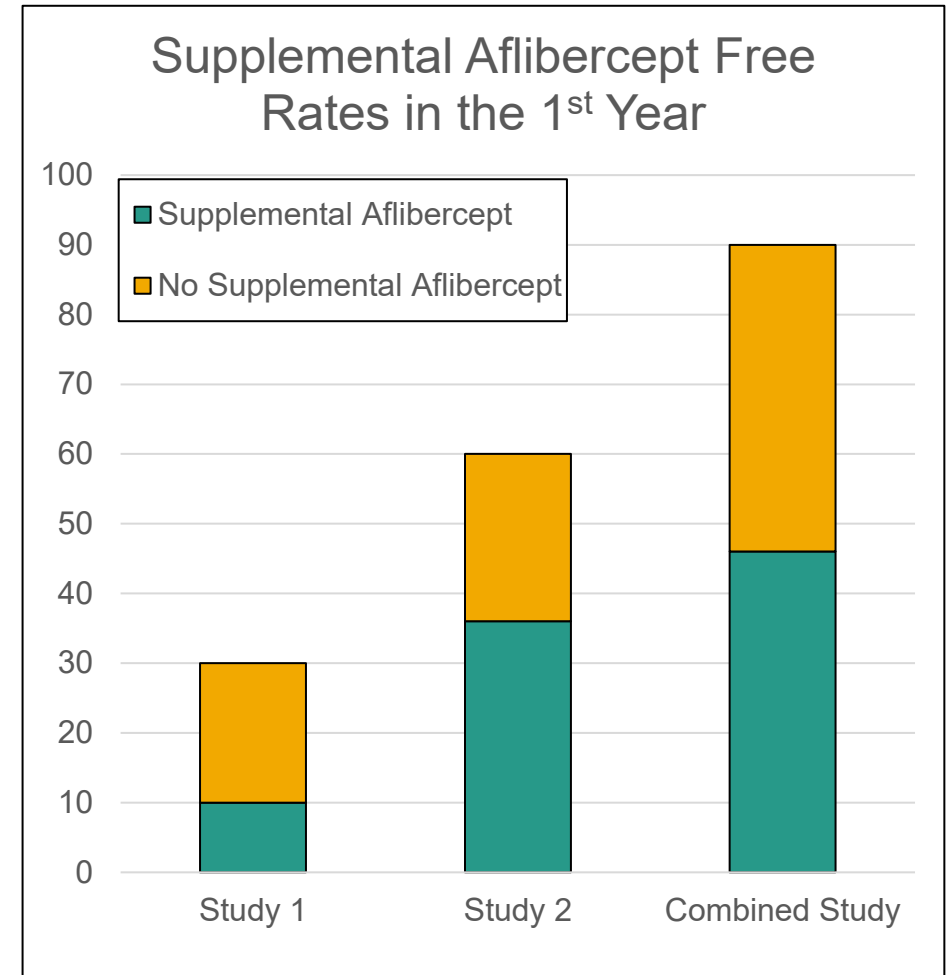
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Training Set

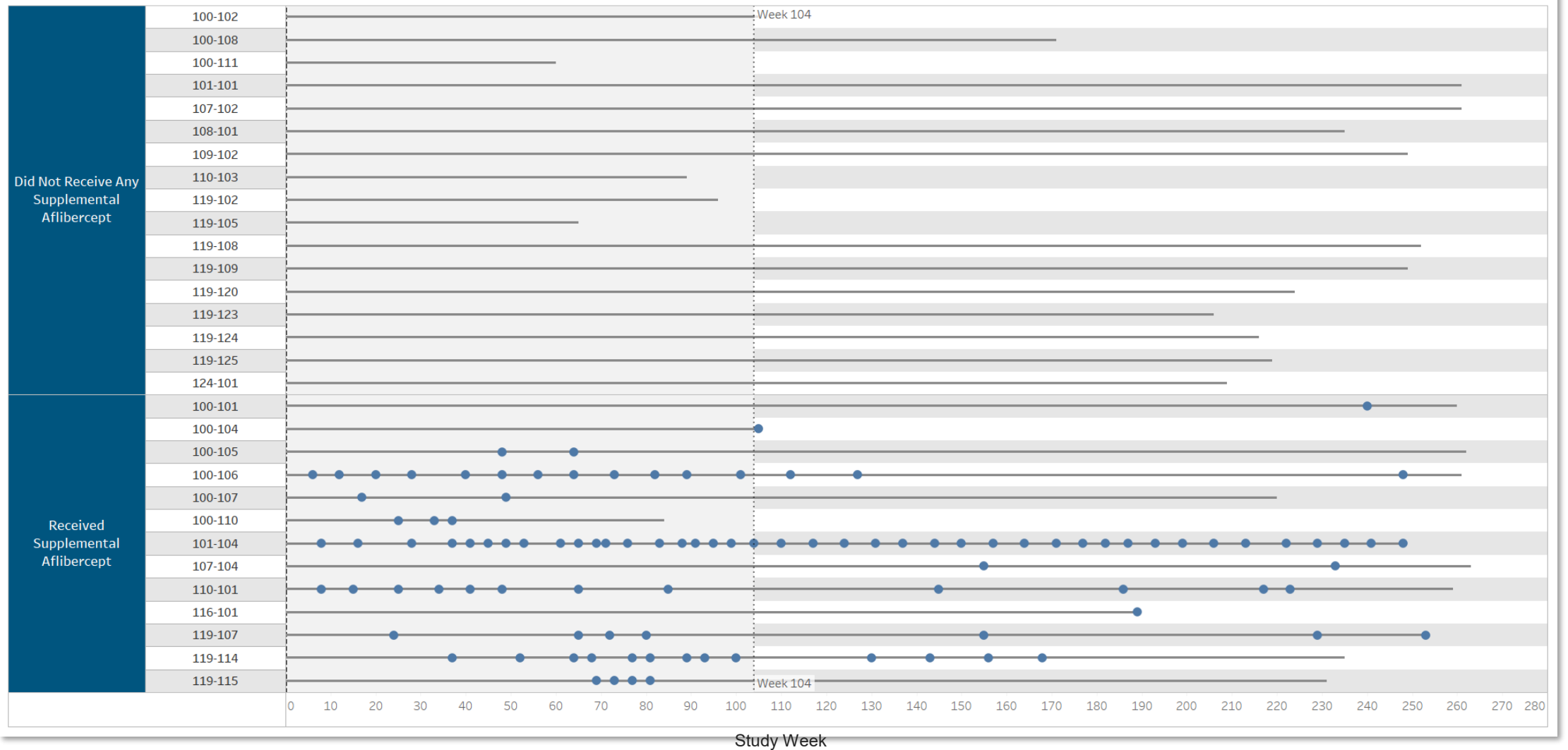


Testing Set

- Build a proof-of-concept ML pipeline that could predict supplemental aflibercept use based on Study 1 and Study 2 synthetic clinical data that mimics OPTIC and LUNA
- Supplemental Aflibercept → addition Aflibercept treatments given to participants due to vision loss, increase in Central Subfield Thickness (CST), or new/worsening hemorrhage
- **End Goal:** Identify participants to monitor based on patterns from previous studies in Study 3 (which mimics ARTEMIS)
- Working with a 50/50 split with the data makes this end target more promising for ML applications



Adverum OPTIC and OPTIC EXT Supplemental Injection Swim Lane Plots



Identify similar datapoints present in all three studies



Converting into usable dataset by joining and cleaning data sources



Create usable tables with consistent data

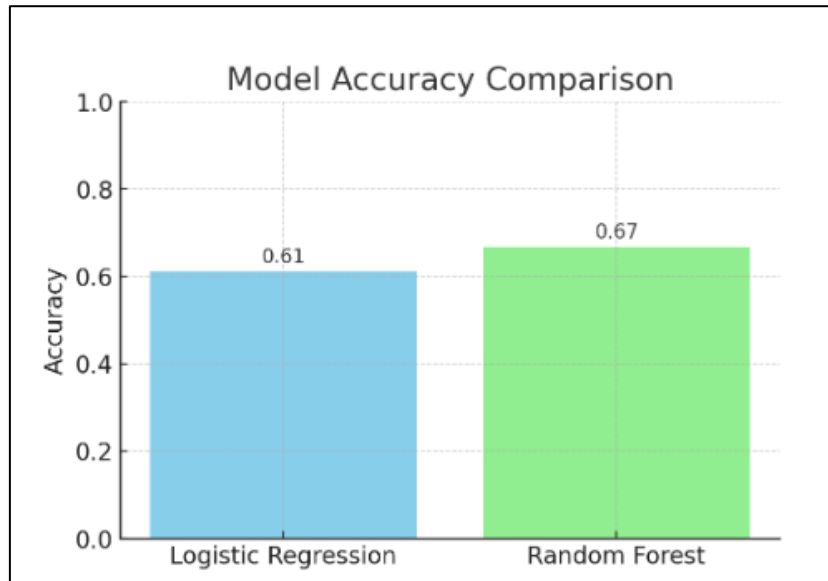
Study 1 + Study 2

Study 3 w/ matching columns

- The synthetic consolidated data contained a total of 90 subjects with both demographic and lab data
- Split the Study 1 + Study 2 combined data into separate training and testing sets
 - Training set (80% of the data)
 - Testing set (20% of the data)
- Run various types of algorithms using the training set to evaluate against the testing set
- **Reason:** Get an informed idea on which model would be best at predicting Supplemental Aflibercept use on Study 3 participants by 52 weeks

```
USER$MRAJA.PUBLIC ▾ Settings ▾  
  
1  -- Rescue Aflibercept Retreatment  
2  LEFT JOIN (  
3    SELECT  
4      PATIENT_ID,  
5      MAX(EXYN) AS EXYN  
6    FROM STUDY1.SUPPLEMENTAL_AFLIB  
7    GROUP BY PATIENT_ID  
8  ) RETRT  
9    ON RETRT.PATIENT_ID = ADM.PATIENT_ID  
10  
11  -- Lab Results (screening only)  
12  LEFT JOIN LAB_RESULTS LAB  
13    ON LAB.SUBJECT_NUMBER = ID.SUBJECT_ID  
14    AND LAB.VISIT_NAME = 'SCREEN'  
15  
16  
17  GROUP BY  
18    -- Patient Identifiers  
19    PATIENT_DISPLAY_ID_FULL,  
20    PATIENT_ID,  
21    ID.SUBJECT_ID,
```

- Model Performance from Simulated Tests:
 - Logistic Regression → 61% Accuracy
 - Random Forest Classifier → 67% Accuracy
- Developed a model that predicts if participants will need Supplemental Aflibercept treatments



```
jupyter ML_RA_PREDICTION Last Checkpoint: 6 seconds ago
File Edit View Run Kernel Settings Help
+ ✂ 📄 📋 ▶ ■ ↺ ⏩ Code ▼

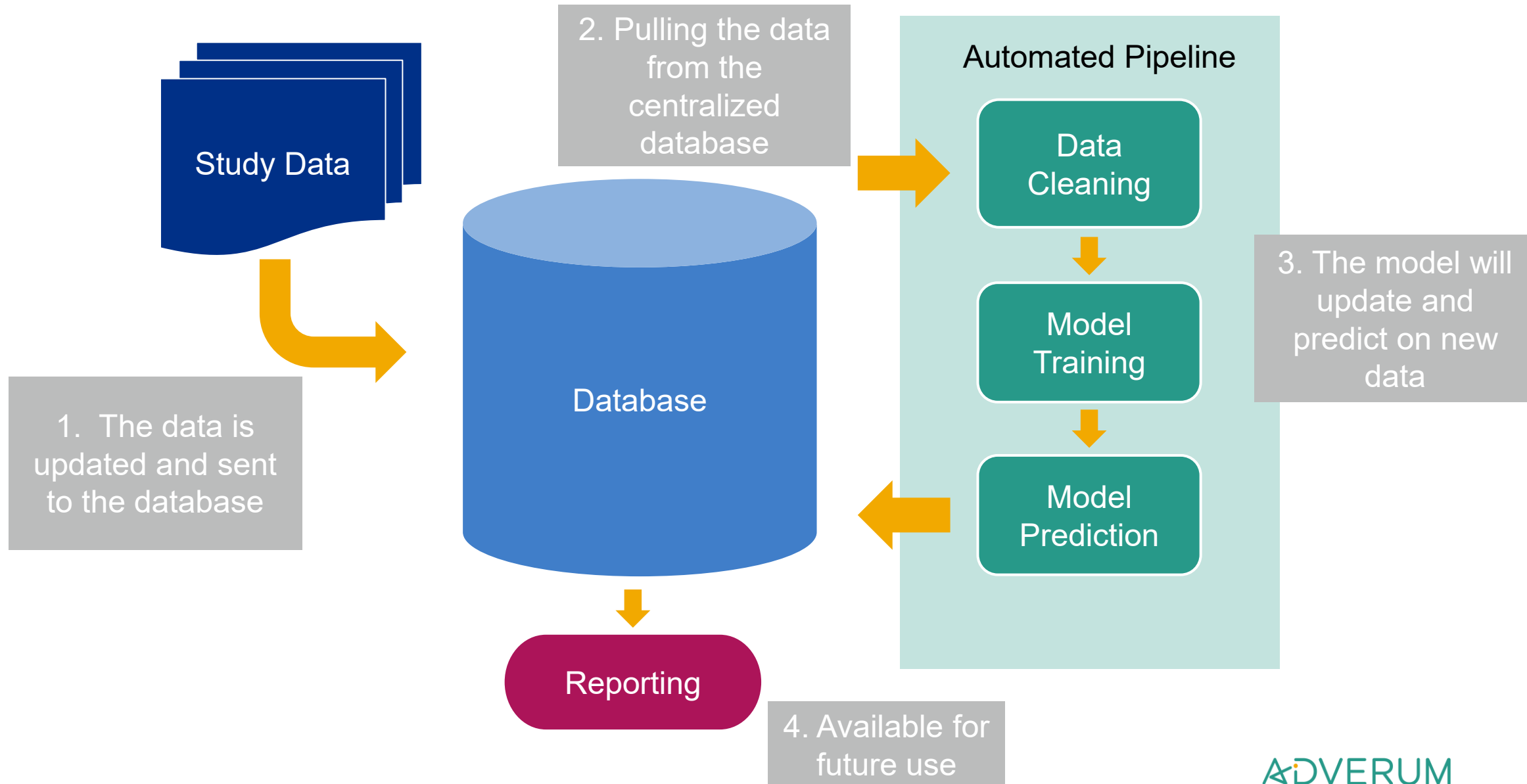
•[2]: # Import libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, con
import joblib

# Load dataset
df = pd.read_csv(r".\Study1+Study2_combined_dataset.csv")

# Encode target variable
df['RESCUE_AFLIBERCEPT'] = df['RESCUE_AFLIBERCEPT'].astype(str).str.up
df = df[df['RESCUE_AFLIBERCEPT'].notna()]

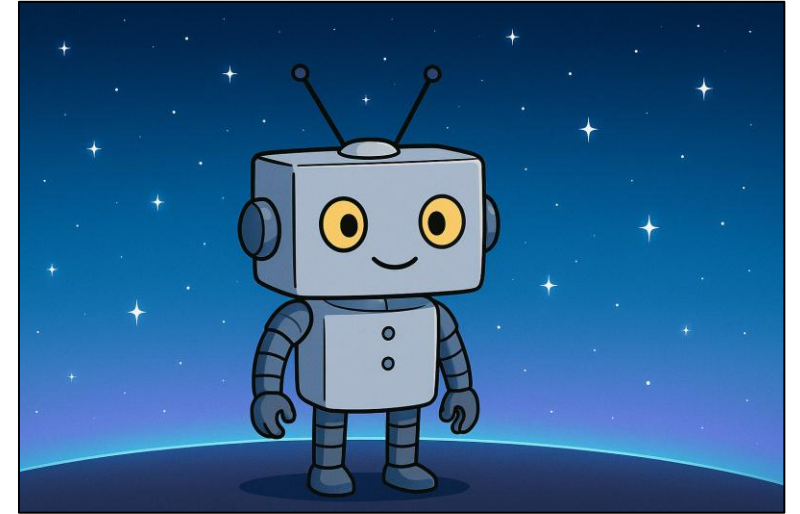
# Store SUBJECT info
subject_ids = df['SUBJECT']

# Separate features and target
target = 'RESCUE_AFLIBERCEPT'
features = df.columns.drop([target])
```



Limitations:

- ML Models are limited by the volume of data in the training set
 - Our synthetic data mimics the volume of available datasets



Future Steps:

- Deploy this model using real-world data
 - Repeat process with more training sets to increase accuracy
- Monitor participants with a high predictability of needing supplemental aflibercept
- Use ML Models to predict on different endpoints
- Analyze correlations which could inform screening/prescreening efforts in future studies

- Special thanks to the Medical Affairs team (especially Phoebe and Jeff) for guiding me through the process as well as the other interns + the People team
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